

Forecast-Aware Rolling-Horizon Optimization for Hybrid Wind-Storage Dispatch

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Summer 2026 REU | Pyron wind farm case study | ERCOT price data | PNNL CAES assumptions

Main takeaway

Hourly uncertainty-aware dispatch gave the strongest COVE reduction while respecting storage limits.

1. Problem and Goal

- Wind farms do not control when wind arrives.
- Electricity prices change hour by hour.
- Storage can hold wind energy and release it later.
- The goal is to choose direct delivery, charging, discharging, and curtailment to increase value.
- The hard part is making decisions before the future is known.

Main comparison: a 100 MW constant-output CAES benchmark. It is a realistic storage rule, while wind-only is secondary context.

Complete Forecast-Aware Dispatch Pipeline



Figure 1. End-to-end pipeline: forecast wind/price, optimize dispatch with Gurobi, then score realized revenue and COVE.

2. Data and Storage Setup

- Dataset: Pyron wind farm, 1980-2023 hourly records.
- Forecast training: historical data before 2014.
- Testing period: 2014-2023 unseen years.
- Price: ERCOT locational marginal price.
- Storage: CAES, 100 MW, 10 h, 1000 MWh capacity.
- SoC bounds: 200-1000 MWh; initial SoC: 600 MWh; RTE: 55%.
- All dispatch results carry SoC forward chronologically.

Common CAES Constraints Used Across Current Runs

Constraint	Value
Storage power	100 MW charge / 100 MW discharge
Capacity	1,000 MWh
SoC bounds	200-1,000 MWh
Initial SoC	600 MWh
RTE	55%, discharge side
Grid cap	249 MW
Grid charging	Not allowed

Figure 2. Dispatch is constrained by wind-only charging, grid export, charge/discharge limits, and chronological SoC.

3. Forecasting

- Causal lag/ridge uses only past information.
- It predicts generated power directly, not wind speed.
- Best forecast RMSE: 21.24 MW.
- This forecast feeds the realistic dispatch experiments.
- Other tested methods included persistence, speed-power curve, RNN, physics, and probabilistic outputs.

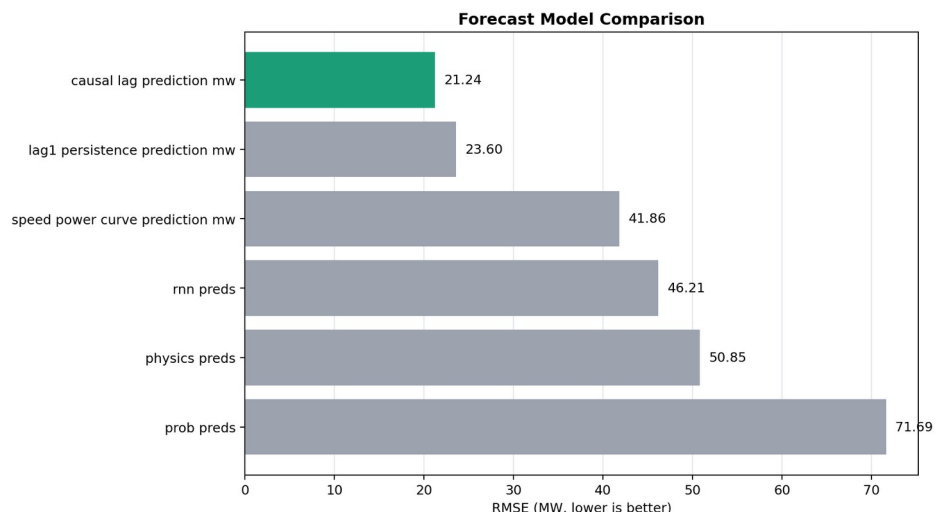


Figure 3. Causal lag/ridge had the lowest power-prediction RMSE among tested forecast outputs.

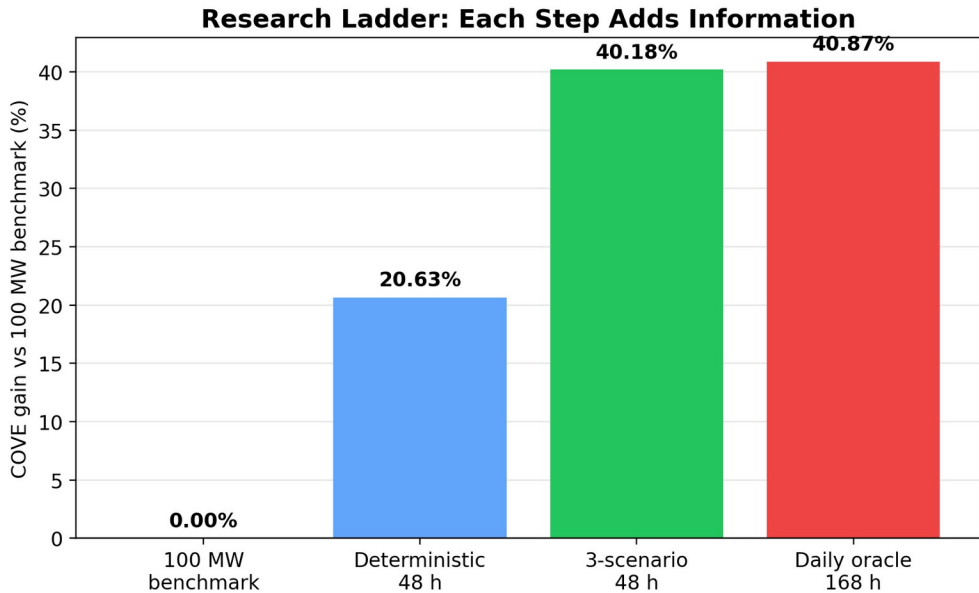


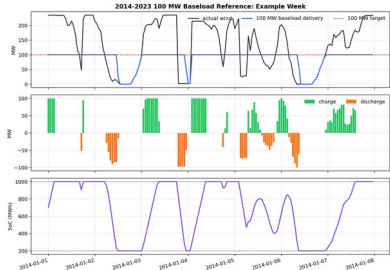
Figure 4. Research ladder: start from 100 MW benchmark, add forecasting, add rolling-horizon dispatch, add scenarios, then compare to oracle.

4. How the Optimizer Decides

Gurobi solves a mixed-integer linear program. It is the mathematical planner that chooses a feasible storage schedule under the rules below.

- P_direct: wind sent straight to the grid.
- P_charge: wind stored in CAES.
- P_discharge: stored energy released to the grid.
- SoC: storage energy carried forward through time.
- Curtailment: wind not used because of physical limits.

Rolling horizon means: plan ahead, execute the near-term action, update the battery, then solve again. Revenue uses realized delivered power times realized price. COVE is lower when delivered energav is more valuable.



Stage	Main idea	Best result vs 100 MW
Benchmark	100 MW rule-based CAES	reference
Forecast	Causal lag/ridge predicts wind power before dispatch	RMSE 21.24 MW
Rolling horizon	Gurobi uses one predicted future and replans	48 h: 20.63% COVE gain
Scenarios	Several futures are optimized together	3 scenarios: 40.18% COVE gain
Oracle	Perfect future wind and price; not deployable	168 h: 40.87% COVE gain

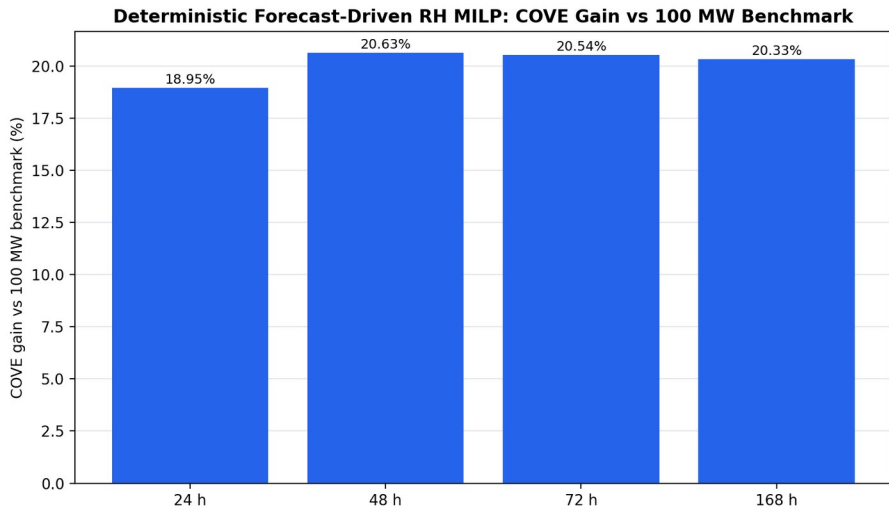


Figure 5. Deterministic dispatch found that 48 h lookahead gave the best COVE gain under the 100 MW benchmark.

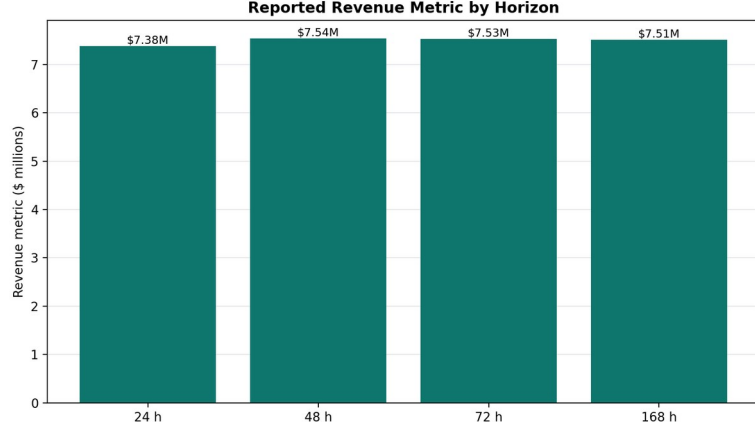


Figure 6. Longer horizons do not automatically win when forecasts are imperfect; 48 h balanced planning value and forecast error.

5. Key Results

Primary comparison: 100 MW constant-output CAES benchmark. Wind-only is side context. Higher COVE gain means the storage policy makes energy more valuable per cost.

21.24 MW

Best forecast model

causal lag/ridge RMSE

20.63%

Best deterministic dispatch

48 h COVE gain vs 100 MW

40.18%

Best scenario controller

3 scenarios, hourly replan

40.87%

Oracle upper bound

168 h perfect future

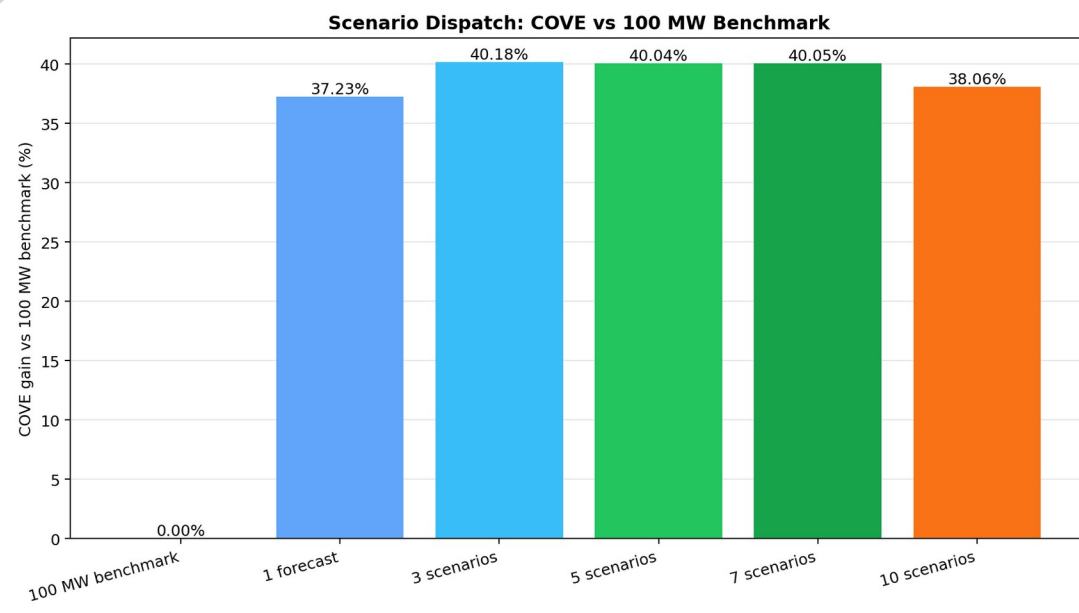


Figure 7. The hourly uncertainty-aware controller worked best with 3 scenarios; too many scenarios became conservative.

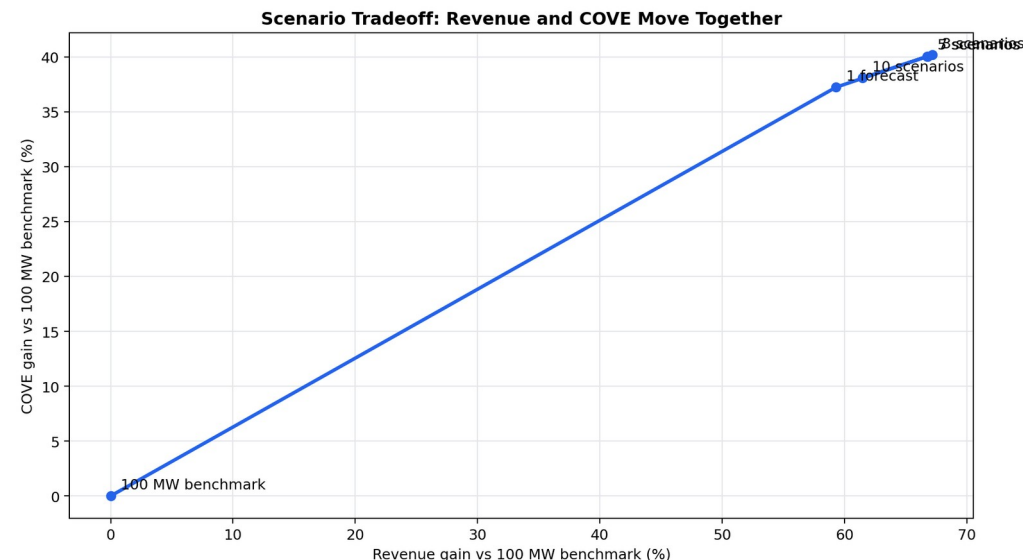


Figure 8. The best scenario case had the highest revenue and lowest COVE among scenario counts.

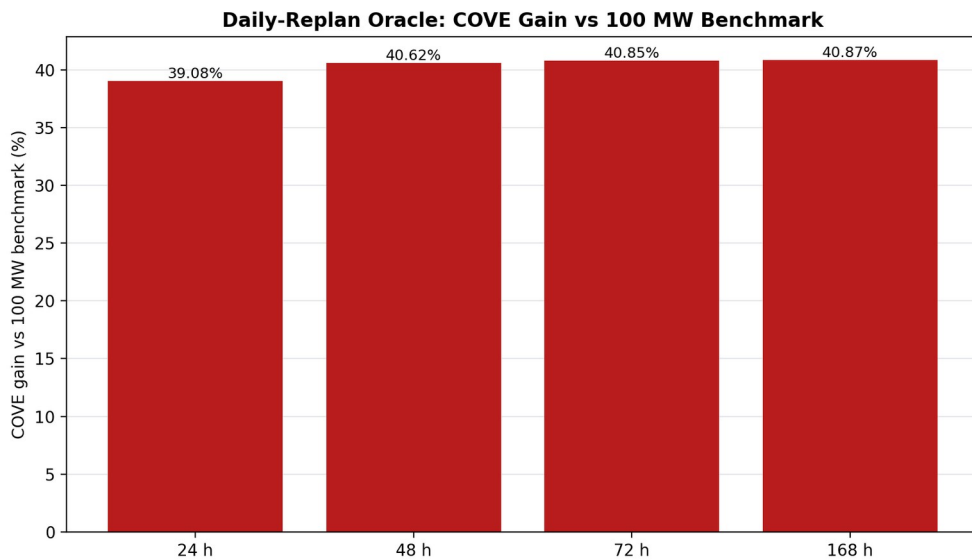


Figure 9. Oracle dispatch is not realistic because it knows the future, but it shows the ceiling for the controller.

6. Main Conclusion

- Forecast quality matters, but dispatch logic matters too.
- A 48-hour deterministic lookahead was best when one forecast was used.
- Hourly uncertainty-aware replanning improved the best result because it corrected mistakes quickly.
- Three scenarios were enough to capture useful uncertainty without making the optimizer overly cautious.
- The project shows forecasting and optimization must be designed as one connected control system.

Practical message: hybrid wind-storage control should tune the forecast horizon, replanning frequency, storage assumptions, and scenario count together, not separately.

Definitions for the reader

- 100 MW benchmark: rule-based storage case that tries to deliver 100 MW every hour.
- Wind-only: no storage; actual wind goes to the grid up to the 249 MW cap.
- Rolling horizon: solve the future, execute the near-term action, then solve again.
- Scenario dispatch: solve against several possible futures instead of one forecast.
- Oracle: perfect-future case used only as an upper bound.
- COVE: cost divided by valued delivered energy; lower is better.
- Constraint violation check: confirms storage and grid rules were not broken.

Acknowledgements

Thank you to Dr. Chris Qin, Nora Hosseiniimemi, Zach Lawrence, Jessica Yao, and the WSU Summer Research Program. Data and assumptions draw from the Pyron wind-storage dataset, ERCOT LMP records, and PNNL CAES storage references.

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